

AI Engineer Job-Ready Course with Internship & Placement Assistance

Learn PYTHON, FastAPI, NLP, GEN-AI, RAG, AGENTIC-AI, MLOPS & AWS

Website - www.aidisha.in

Duration- 5 Months.

Phone- [+91 8010125091](tel:+918010125091)



- Industry based intership on real world project
- Lots of Practical exercises
- 5 + hands on projects
- Interview Preparation
- Resume building,
- Mock Interview

This course is highly curated and uniquely designed according to the latest industry standards. You will learn all the stack required to work in Artificial Intelligence with real-time industry projects.

Learn PYTHON



Python has become the dominant language for Artificial Intelligence (AI) due to its readable, English-like syntax and a vast ecosystem of pre-built libraries that allow developers to focus on solving problems rather than writing complex low-level code.

Detailed Syllabus:

1 Introduction to Python

2 Getting Started

3 Python Basics

4 Strings

5 Decision Control Instruction

6 Repetition Control Instruction

7 Console Input/output

8 Lists

9 Tuples

10 Sets

11 Dictionaries

12 Comprehensions

13 Functions

14 Recursion

15 Functional Programming

16 Modules and Packages

17 Namespaces

18 Classes and Objects

19 Intricacies of Classes and Objects

20 Inheritance in Python

21 Iterators and Generators

22 Exception Handling

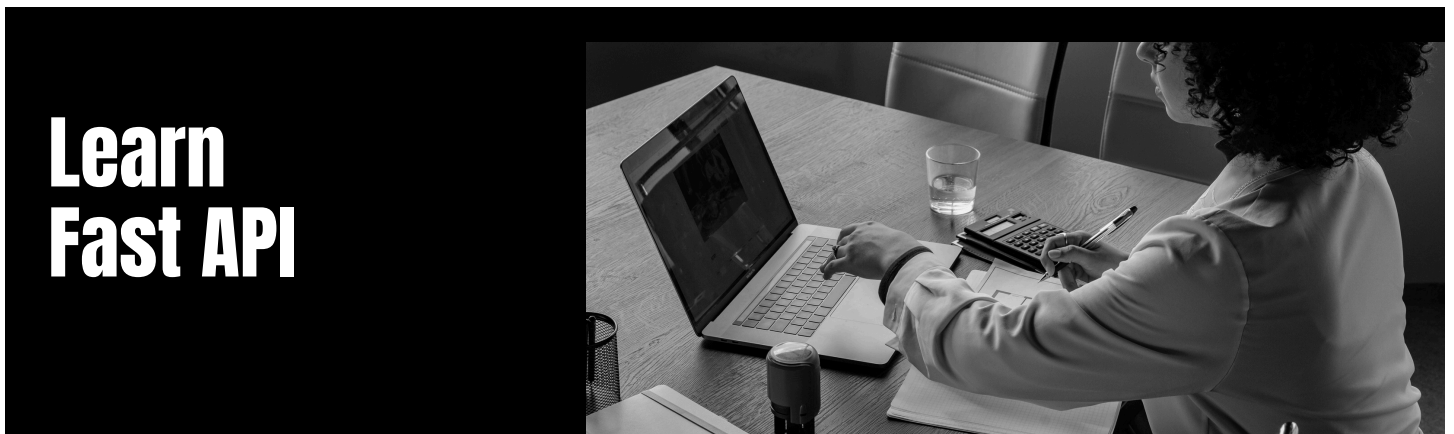
23 File Input/Output

24 Miscellany

25 Concurrency and Parallelism

26 Synchronization

We'll dive into a variety of programs for every topic! ✨



FastAPI is a modern, fast (high-performance), web framework for building APIs with Python based on standard Python type hints.

Detailed Syllabus:

1. What is an API? | Introduction to APIs

2. HTTP Methods in FastAPI

3. Path & Query Params in FastAPI

4. Pydantic Python library for data validation
5. Post, PUT, Delete Request in FastAPI
6. FastAPI + Docker
7. Deploy FASTAPI API and Postman testing

We'll dive into a variety of programs for every topic! ✨

Learn ML & Natural Language Processing



Natural language processing, or NLP, combines computational linguistics—rule-based modeling of human language—with statistical and machine learning models to enable computers and digital devices to recognize, understand and generate text and speech.

Detailed Syllabus:

1. **Essential ML Topics for an interview**
2. **Introduction to NLP**
3. **End to End NLP Pipeline**
4. **Text Preprocessing Techniques**
5. **Text Representation Techniques**
6. **Word2Vec**
7. **CBOW**

- 8. Skip-gram
- 9. NLP Project

We'll dive into a variety of programs for every topic! ✨

Learn Generative AI & Agentic AI



Generative artificial intelligence (AI) is a type of AI that generates images, text, videos, and other media in response to inputted prompts.

Detailed Syllabus:

1. Introduction to Generative AI

1. What is generative AI?
2. Why are generative models required?
3. Understanding generative models and their significance
4. Recent advancements and research in generative AI
5. Generative AI end-to-end project lifecycle
6. Key applications of generative models

2. Introduction to Large Language Models

1. Application and use cases of LLMs
2. Transfer learning in NLP
3. How to use pre-trained transformer-based models
4. How to perform finetuning of pre trained transformer-based models

3.Introduction to Hugging face and its applications

1. Why the need for a hugging face?
2. Introduction of Hugging Face Transformers
3. Hugging face API key generation
4. Hugging Face Transfer learning models based on the state-of-the-art transformer architecture

4.Guide to Open AI and its Ready to Use Models with Application

1. Introduction to OpenAI
2. What is OpenAI API and how to generate OpenAI API key?
3. Installation of OpenAI package
4. Experiment in the OpenAI playground
5. How to setup your local development environment
6. Different templates for prompting
7. OpenAI Models GPT-3.5 Turbo DALL-E 2, Whisper, Clip, Davinci and GPT-4 with practical implementation

5.Prompt Engineering Mastering with OpenAI

1. Introduction to Prompt Engineering
2. Different templates for prompting
3. Prompt Engineering: What & Why?
4. Prompt Engineering & ChatGPT Custom Instructions
5. The Core Elements Of A Good Prompt
6. Which Context Should You Add?
7. Zero- One- & Few-Shot Prompting
8. Ask-Before-Answer Prompting
9. Perspective Prompting
10. Contextual Prompting
11. Emotional Prompting
12. Laddering Prompting
13. Using ChatGPT For Prompting
14. Find Out Which Information Is Missing
15. Self-evaluative Prompting

16. ChatGPT-powered Problem Splitting
17. Reversing Roles
18. More Prompts & Finding Prompt Inspirations
19. Super Prompts Like CAN & DAN

6.Vector database with Python for LLM Use Cases

1. Introduction to vector database
2. Vector database foundation
3. Vector database use cases
4. Text embedding
5. Vector similarity search
6. Storing and retrieving vector data in SQLite
7. Chromadb local vector database part1 setup and data insertion
8. Query vector data
9. Fetch data by vector id
10. Database operation: create, update, retrieve, deletion, insert and update
11. Application in semantic search
12. Building AI chat agent with langchain and openai
13. Weviate Vector Database
14. Pinecone Vector Database

7.Hands-on with LangChain (Gen-AI)

1. Introduction to LangChain
2. Prompts in LangChain
3. Models in LangChain
4. Structured Output in LangChain
5. Output Parsers in LangChain
6. Chains in LangChain
7. Runnableables in LangChain
8. Document Loaders
9. Text Splitter
10. Embedding
11. Vector db

12. Retriever
13. Tool Calling
14. Retrieval-Augmented Generation (RAG) using Langchain components

8.Hands-on with LangGraph (Agentic-AI)

1. Introduction to LangGraph and AI Agents
2. Core Elements of LangGraph - state, nodes, edges
3. Building Your First AI Agent
4. Introducing Memory in AI Agents - Short-Term Memory, Memory Across Multiple Sessions
5. Advanced Routing and Customization of AI Agents- Conditional Routing, Streaming, External API Integrations
 - Weather API Integration: A Step-by-Step Example
 - Dynamic API Integration with User Input
 - Calculator API Integration
6. AI Agent Architectures - ReAct
 - 10 Examples of ReAct Agents in LangGraph
7. Human-in-the-Loop Agents, Incorporating Human Feedback for Smarter
 - Implementing Simple Breakpoints, Decision-Making- Breakpoints, Checkpoints, and State Editing
 - ReAct Agent Example with Financial Stock Purchase Use Case
 - ReAct Agent with Human Input
8. Plan-and-Execute Agents:
 - IT Troubleshooting and Diagnostics Agent
 - Practical Example 2. Business Workflow Automation Agent
 - Practical Example 3: Customer Support Chatbot
9. Agentic Retrieval-Augmented Generation (RAG) in LangGraph
10. Advanced RAG Architectures (Self-RAG, Corrective RAG and Adaptive RAG)
11. Multiagent Architectures
 - Agent Supervisor
 - Practical Example 1: A Research Assistant Network
 - Practical Example 2: Customer Service Automation
 - Practical Example 3: Stock Analysis
 - Practical Example 4: Portfolio Management
12. Hierarchical Agent Teams
13. Specialized Agent Architectures
 - Event-Driven Agents

- **Cognitive Architectures**
- **Model-Based Agents**

14. Testing AI Agents

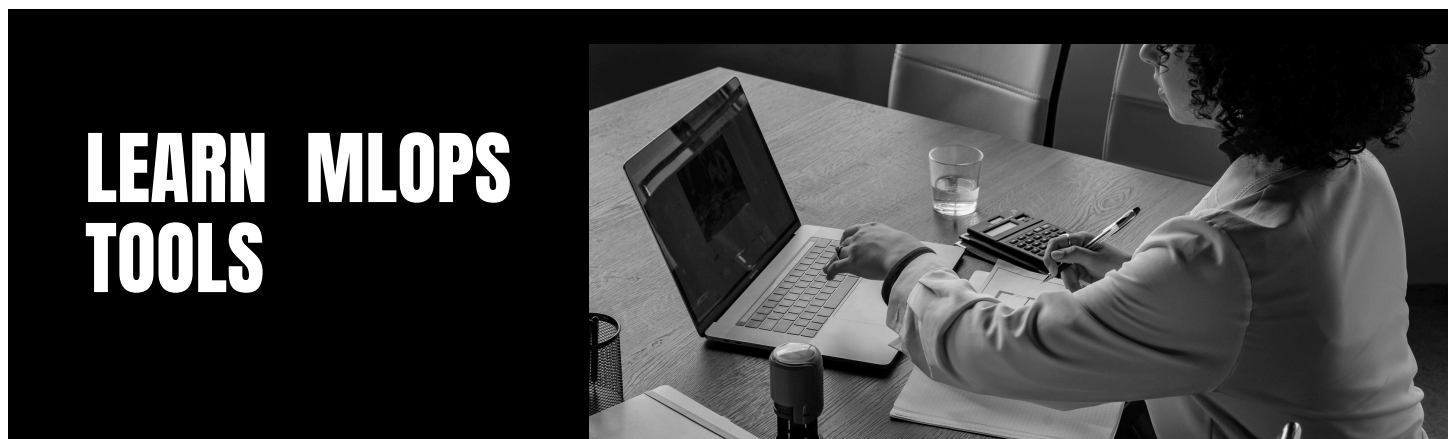
15. Frontend Development for LangGraph-Powered AI Agents

16. Testing AI Agents with Test-Driven Development (TDD)

17. Deploying AI Agents into Production

9.End to End Projects (Industry Level Projects)

- 1. Project1: RAG with END to END deployment**
- 2. Project2: Fine Tuning with END to END deployment**
- 3. Project3: Agentic AI with RAG with END to END deployment**
- 4. Project4: Multi-agent Systems with END to END deployment**
- 5. Project5: Multi-agent Systems & MCP with END to END deployment**



Detailed Syllabus:

1.Linux

2.Git and Github

3.Docker

4.CICD Pipelines

5.Ansible

6.Terraform

7. Kubernetes - AWS EKS

LEARN AWS SERVICES



Certification:

- 1.aws certified solutions architect – associate**
- 2.aws certified generative ai developer – professional**

Detailed Syllabus:

- 1.Ec2**
- 2.VPC**
- 3.DNS - Route 53**
- 4.Load Balancer**
- 5.Auto Scaling**
- 6.IAM**
- 7.Simple Storage Service**
- 8.Databases**
- 9.Lambda**
- 10.ECS**
- 11.EKS**

12.Amazon Bedrock

13.Amazon Sagemaker

14.Amazon Bedrock Agentcore

PRACTICE QUESTIONS



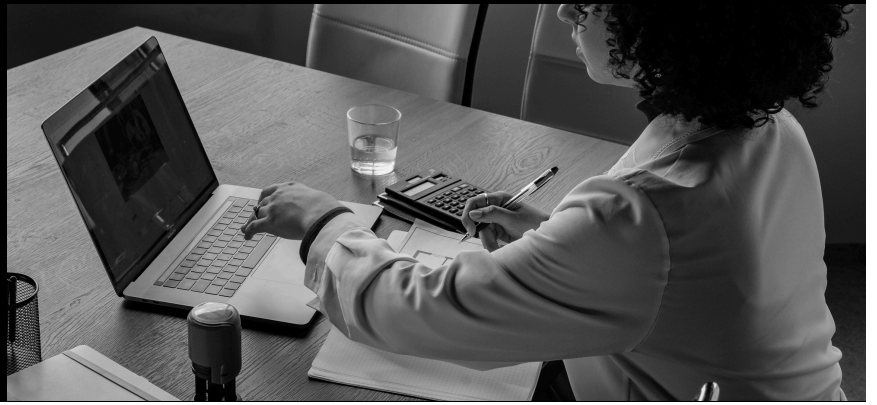
We provide practice questions for every topic covered in the program. Students will receive 1,000+ practice questions throughout the course to strengthen their understanding and improve problem-solving skills.

INTERVIEW QUESTIONS



We provide topic-wise interview questions to help students prepare for technical discussions and assessments. Students will receive 1,000+ interview questions throughout the program.

PYTHON PROGRAMS



Students will work on 500+ Python programming exercises and interview-oriented coding problems commonly asked in L1 and L2 technical interviews. These programs help build strong coding skills and increase confidence in clearing technical rounds.

RESUME BUILDING



We provide dedicated resume-building support that meets current industry standards and helps improve students' chances of getting shortlisted for interviews.

NAUKRI, LinkedIn, GITHUB PROFILE BUILDING



Throughout the program, students build a strong GitHub portfolio by regularly committing projects. We also help optimize Naukri and LinkedIn profiles with the right keywords, skills, projects, and certifications to improve recruiter visibility and increase interview opportunities.

MOCK INTERVIEWS



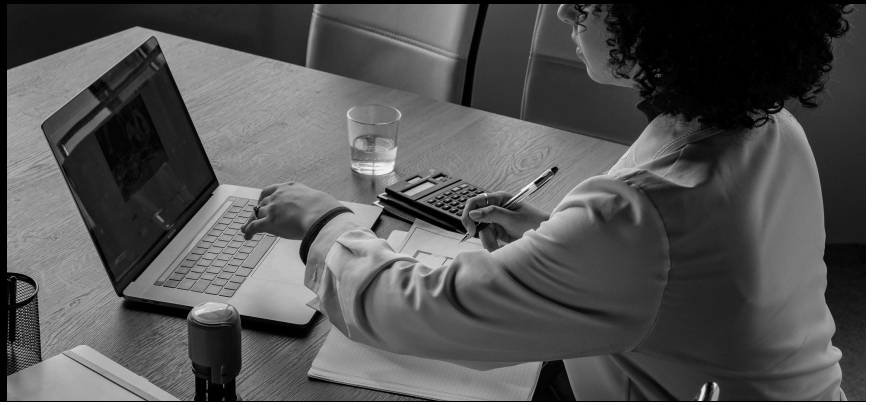
Every Sunday, students participate in mock interviews based on the topics covered during the program. These interviews are designed to simulate real industry interview experiences, helping students build confidence and improve their communication, problem-solving, and technical skills. Detailed feedback and improvement suggestions are provided after each interview to help students prepare effectively for actual job interviews.

Industry-Based Internship on Real-World Projects



Gain hands-on experience through an industry-based internship where you will work on real-world AI projects and development workflows. During the program, students will build 5 end-to-end capstone projects covering Python, NLP, Generative AI, RAG, Agentic AI, MLOps, and AWS. This practical experience helps students develop job-ready skills, strengthen their portfolios, and gain the confidence to work on real industry projects.

Eligible JOB Roles



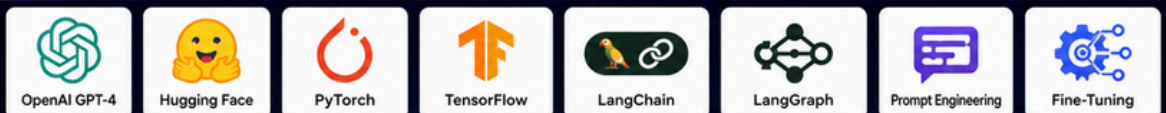
1. Generative AI Engineer
2. AI Engineer
3. Machine Learning Engineer
4. LLM Engineer
5. RAG Engineer
6. Agentic AI Engineer
7. NLP Engineer
8. Python Developer
9. AI Application Developer
10. AI/ML Developer
11. DevOps/MLOps Engineer
12. Prompt Engineer
13. Applied AI Engineer
14. AWS Engineer

TOOLS YOU WILL MASTER

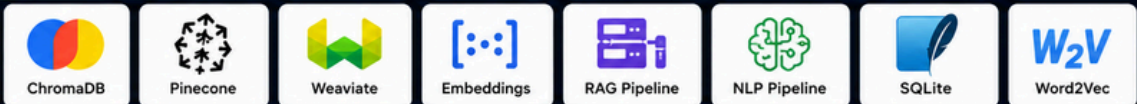
1. PROGRAMMING & BACKEND



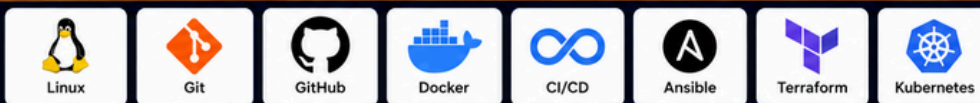
2. GENERATIVE AI & LLMs



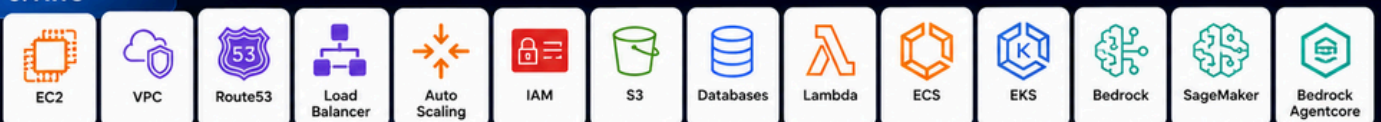
3. RAG, NLP & VECTOR DBs



4. MLOPS



5. AWS



Our Placement Partners

